

Foreign Ownership Restriction and Momentum

— Evidence from Emerging Markets

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Abstract

This paper finds evidence in support of the profitability of momentum strategies in emerging markets. More interestingly, this paper finds that, for stocks with different level of foreign ownership restriction, the profitability of momentum strategies is significantly different. In particular, the momentum strategy on highly investible stocks significantly outperforms the strategies on non-investible stocks. This paper also shows that world market risk, size, turnover, country-specific factors, as well as foreign investors' feedback trading behaviour and their overconfidence psychological bias cannot fully explain the higher profitability of the momentum strategies on stocks with higher foreign ownership. The results are quite consistent with Hong and Stein (1999)'s gradual information diffusion model, suggesting that high foreign investor participation might delay the information transmission among all investors.

1. Introduction

Since the mid-1980s, the majority of emerging markets have started liberalizing their equity markets and allowing foreign investors to participate. Following the market liberalization, there has been a surge of studies on the impact of liberalization on the economy in many aspects. For instance, some papers show that liberalization has had a positive effect on the economy of emerging markets via lowering the cost of capital, increasing the prices as well as increasing the private physical investment (Bekaert and Harvey, 2000; Henry, 2000; and Kim and Singal, 2000). Other researches focus on the volatility of stock markets after liberalization and document mixed findings¹. However, little attention has been paid to the impact of market liberalization on the predictability of stock returns. On the other hand, there have been some papers studying the profitability of various investment strategies in an international emerging markets context. However, one key issue that has never been addressed is the feasibility of these strategies. This issue is of particular importance and interest when emerging markets are involved because not all stocks in emerging markets are accessible to international investors. Even after the market liberalization, most emerging markets still impose limitations on level of foreign ownership to securities. And the level of constraint varies across countries, industries and securities. Therefore, one objective of this study is to investigate the impact of foreign ownership restriction on the predictability of stock returns in emerging markets. In particular, we focus on the relationship between foreign ownership restrictions and profitability of momentum strategies.

¹ Shamila Jayasuriya (2002) summarizes studies on stock market liberalization and return volatility.

After the restriction on foreign capital is removed, foreign investors are attracted to emerging markets for at least two reasons: First, compared with those at developed market, prices of stocks at emerging markets are relatively low and expected returns are high (Harvey, 2000). Second, the correlation of emerging markets with world market returns has been historically low or even negative (Harvey, 1995). Even though the volatility of a single emerging market is high, investing into emerging markets can diversify the volatility of a world portfolio. For either reason, when foreign investors pour into emerging markets, they may have impact on the markets' behaviour.

There have been many researches on foreign investors, mostly focusing on information asymmetry, i.e., whether foreign investors are better informed or less informed relative to domestic investors. Answers to this question remain controversial. Many researchers believe that foreign investors are generally less informed than domestic investors. For instance, Choe, Kho and Stulz (2000), Hau (2001) and Dvorak (2005) document that local investors outperform foreign investors in Korea, German and Indonesia stock markets. Ghargh (1993), Brennan and Cao (1997), and Kang and Stulz (1997) attribute the concentration of portfolio investment in domestic assets (known as "home bias") to informational disadvantage of foreign investors caused by linguistic and cultural barriers. Grinblatt and Keloharju (2001) using Finnish data provide evidence on the linguistic, cultural and geographic barriers for foreign investors. Coval and Moskowitz (1999) show more evidence on information asymmetry even in the absence of cultural and linguistic information barriers by showing that U.S. domestic portfolio funds are geographically biased toward the home of the fund.

Previous researchers apply the results from these studies in explaining home bias, the size and movement of international portfolio flows and difference in valuation by foreign and domestic investors. However, one question people have missed is that the information asymmetry between investors could also affect the predictability of stock returns. Holden and Subrahmanyam (1992) and Foster and Viswanathan (1993) show that share prices will reflect new information more rapidly as the number of informed investors increases. In other words, information diffuses more slowly as uninformed investor increases. If foreign investors are not as well informed as domestic investors, their participation in emerging markets will retard the diffusion of information among investors and the incorporation of the information by stock prices. Hong and Stein (1999) gradual-information-diffusion model implies that momentum profits will be more pronounced for stocks with slow information diffusion. According to the prediction of this model, we conjecture that momentum is more pronounced among stocks with high foreign ownership.

Despite the emphasis and abundant evidence on the information disadvantage and simple trading behaviour of foreign investors, there is also some empirical evidence supporting the opposite opinion that foreign investors are better informed and trade on information. Grinblatt and Keloharju (2000), Seasholes (2000), and Karolyi (2002) document that foreign investors seem to outperform domestic investors in Finnish, Taiwan and Japanese stock markets. Froot and Ramadorai (2001) look at cross-border equity flows from State Street and show that foreign purchases can predict prices in foreign markets. However, the short-term price performance of trade by foreign and domestic investors may not be a good indicator of their information advantage or

disadvantage. Grinblatt and Keloharju (2000, 2001) also find that foreign investors tend to be momentum investors. Brennan and Cao (1997)'s theoretical model and empirical evidence suggest that foreign investors should pursue momentum strategies because they are less informed than domestic investors. In fact, even if foreign investors tend to be professional investors, and they are better at collecting, filtering and processing useful information from the market, the cultural, linguistic and geographic barriers would inevitably delay such process and cause the information to diffuse gradually across all the investors, which will cause underreaction in stock prices and momentum in returns.

Thus, we expect to see that for stocks that are more open to foreign investors, or have larger foreign ownership, information diffuses more slowly among investors. According to Hong and Stein (1999) prediction, such slow spread of information will cause some investors to underreact to the information, and thus induce momentum in stock returns. Whereas, for stocks that are relatively constrained to foreign investors, since ownership is less diversified, information diffuses more quickly and efficiently among domestic investors. Therefore, these stocks are less likely to show momentum.

The variable we use to proxy the openness of the stocks to foreign investors, or the foreign ownership, is the "Foreign Investable Weight Factor (IWF)" from the Standard & Poor Emerging Market Database (EMDB). Foreign Investable Weight Factor is a degree-open factor, which represents the degree to which foreign investors may access the stock². IWF varies substantially across markets and stocks. For all the countries in our sample, the IWF of different stocks varies from 0% to 100%. But the

² If foreign institutions can invest in listed shares, S&P then investigate each security in the S&P/IFCG Index to determine whether there are any corporate by-law, corporate charter, or industry limitations on foreign ownership of the stock. S&P then creates a variable called the "degree of open factor" with value from 0 to 1 that indicates the amount of the security foreigners may legally own (0.00 indicates that the stock is not legally available to foreigners).

mean (or median) IWF of different countries indicates their different degree of openness. For example, some countries like Jordan can be seen as highly closed because the mean IWF is only 9.55% and the median is 0.00%. Other countries like Malaysia, Turkey, Mexico and Argentina, have a mean IWF of about 80% and can be seen as highly accessible to foreign investors. Difference in foreign ownership restriction also gives the third reason why we use emerging market data. It makes it possible for us to investigate the effect of foreign ownership restriction on the profitability of momentum strategies and apply Hong and Stein's gradual-information-diffusion model, by linking investibility or openness of the stock to the information diffusion rate. To do so, we sort stocks into different groups based on IWF, assuming that higher investibility implies higher proportion of foreign ownership and lower information diffusion rate. Our central hypothesis is that stocks with higher IWF should exhibit more pronounced momentum.

Here we have two key assumptions: First is that the degree of investibility indicates the proportion of foreign ownership. Second is that the proportion of foreign ownership indicates the information diffusion rate among investors. In particular, the higher the investibility, the higher the foreign ownership, and the slower the information diffusion rate is. We admit that the investibility factor is not necessarily a good proxy for the degree of foreign ownership. In our empirical tests, we particularly separate stocks that are 100% open to foreign investors from stocks that are partially open to foreign investors and stocks that are strictly non-investable to foreign investors. Though we can not guarantee that investibility factor has a monotonic relationship with actual foreign ownership, there is no doubt that the foreign ownership of stocks that are 100% open to foreign investors is no less than that of stocks that are absolutely non-investable to

foreign investors. Also, the investibility may be correlated with other factors. For example, companies with higher foreign ownership restrictions may tend to be smaller in size, less active in trading, or come from particular countries. Therefore, it is possible that other characteristics of the stocks rather than the foreign ownership restriction may affect the momentum profits. To circumvent this problem, we conduct some robustness tests to separate the effect of investibility from other factors.

The main findings of this study are as follows. Firstly we find evidence in support of the profitability of momentum strategies in emerging markets. Our results indicate that average stock returns are positively related to past performance in short horizon (less than one year). This pattern in stock returns is not limited to a particular market or to stocks with particular features, but is present in eleven out of sixteen emerging markets, and in all stocks with different size or turnover ratio. Our subperiod sample tests indicate that momentum is more pronounced during the 1996-2002 period than during 1990-1995. Secondly, by assuming that the openness of stocks, or the foreign ownership structure of the stocks affects the information diffusion rate, we find that, for stocks with different openness, the profitability of momentum strategies is significantly different. In particular, the momentum strategy on highly investible stocks significantly outperforms the strategies on non-investible stocks. It even outperforms the strategy on US market documented in Jegadeesh Titman (1993). Thirdly, we show that world market risk, size, turnover and country-specific factors cannot fully explain the higher profitability of our momentum strategies on stocks with higher foreign ownership. And finally, consistent with Hong and Stein's underreaction-overreaction prediction, we find that short term

positive autocorrelation in returns turns to be negative at longer horizons, indicating an initial underreaction in stock returns.

The remainder of this paper is organized as follows: Section 2 provides a description of our data, the summary statistics and presents the empirical results on both simple and subsample momentum profits. Section 3 investigates the relation between foreign ownership restriction and momentum profits. It also presents robustness tests on possible explanatory factors of the profits, like world market systematic risk, size, turnover or country specific factors, as well as foreign investors' trading behaviour and psychological biases. Section 4 concludes the paper.

2. Data and Empirical Results

2.1. Data and Descriptive Statistics

All data used in this paper are from the Emerging Markets data Base (EMDB) produced by the International Financial Corporation (IFC). Our sample period runs from January 1990 through July 2002. We limit our sample countries to those with sufficient historical return data, which leaves us 19 countries that the IFC tracks for at least 16 years. In order to test the impact of foreign ownership constraint on the profitability of momentum strategies, we further exclude three countries from our sample: India, Nigeria and Zimbabwe because no fully open stocks are available in these three countries. This filter rule leaves us 16 countries in our sample: Argentina, Brazil, Chile, Greece, Korea, Mexico, Thailand, Jordan, Colombia, Malaysia, Pakistan, Philippines, Taiwan Province of China, Venezuela, Indonesia, and Turkey. We use monthly data on all the stocks in

EMDB from these 16 countries and the data include returns and market capitalization in US dollars, and turnover ratio. Since we use the same source of data as in Rowenhorst (1999), our data inevitably have the same selection bias documented by Rowenhorst (1999)³.

Table 1 presents the descriptive statistics for the sample markets. The first two columns of data are the number of observations and number of stocks sampled in each market. The number of stocks used for each country ranges from 20 (Venezuela) to 264 (Korea). The next six columns report the descriptive statistics for Investibility Weight Factor. IWF varies a lot across countries: Some countries have a high mean value of investibility but low variation such as Turkey (with mean of 82.13% and standard deviation of 29.57%) and Malaysia (with mean of 84.75% and standard deviation of 27.82%). These countries can be considered as highly open⁴. In contrast, some countries have low mean value and low variation such as Taiwan⁵ (with mean of 23.16% and standard deviation of 18.87%) or Thailand⁶ (with mean of 28.96% and standard deviation of 16.48%). These countries are considered to have high restriction to foreign shareholdings. Some countries, like Argentina⁷ (79.13% in mean and 39.11% in standard deviation), have both high mean and variation. While other countries, like Jordan (with mean of 9.55%, and standard deviation of 21.10%) is low in both mean and variation.

³ Rowenhorst (1999) documents that because of the criteria IFC uses to select stocks for its global indices, EMDB bias toward larger and more frequently traded issues.

⁴ EMDB defines these countries as 100% open.

⁵ The market was closed until January 1, 1991. At that time, authorities permitted foreign institutions meeting fairly strict registration requirements to invest in listed stocks, up to a 10% limit of aggregate foreign investment in a firm., The ceiling on inflow was raised to 12% in July 1995, to 15% in September 1995, to 20% in March 1996 and to 30% in November 1996.

⁶ Various Thai laws restrict foreign shareholdings in Thai companies engaged in certain areas of business. For instance, the Banking Law restricts foreign ownership in banks to 25%, the Alien Business Law restricts foreign ownership of stocks in specified sectors to 49% as well. Company by-laws impose restrictions that range from 15% to 65%.

⁷ Argentina is considered 100% open by EMDB. Some corporate limitations apply.

The next 2 columns report the average monthly returns in US dollars for these 16 markets for the whole sample. The average monthly return for these 16 markets is about 0.67% for the whole sample period. To investigate the variation of stock markets over time, we divide our sample period into two sub-periods: January 1990 – December 1995 and January 1996 – July 2002. We notice that the average monthly return is 1.43% for the first subperiod and 0.07% for the second one. The financial crisis decreases the performance of stock markets significantly. It also caused higher volatility in markets: the average standard deviation of the return increases from 18.01% to 20.58%. The size reported in the last three columns of Table 1 is measured as the natural logarithm of the market value of sample stocks (in US dollars). The mean firm size is highest in Taiwan (6.40) and lowest in Jordan (3.17).

2.2. Simple Momentum Strategy

We shall first test the profitability of momentum strategies in the 16 emerging markets. The momentum portfolios constructed are similar to those in Jegadeesh and Titman (1993). In particular, we concentrate on the 6/6 month strategy and sort the sample stocks by their raw returns. The momentum portfolios are constructed as follows: at the end of each month, all sample stocks are sorted based on their past 6-month (ranking period) returns and then assigned to one of five quintiles (P1 represents the worst past performers— defined as losers, and P5 represents the best past performers— defined as winners). The portfolios are equally weighted and held for subsequent 6 months (test period). We then calculate the performance of these five portfolios in the test period. We define the return to a momentum (or zero-net-investment) strategy as the

return on the winner portfolio (P5) less the return on the loser portfolio (P1). Since the trading rules are replicated at the end of each month, overlap in data are generated and the mean returns are the average of all the replications. The t-statistics reported are computed using the Newey-West correction for heteroskedasticity and autocorrelation.

2.2.1 Simple Momentum Strategies on Individual Countries

We first construct the momentum strategies for each sample market for the whole sample period. Table 2 shows the average profits for each country from January 1990 to July 2002. Our results indicate that average stock returns are positively related to past performance. On average, winner portfolios outperform loser portfolios in the test period by 0.58% per month, which is lower than those documented by Jegadeesh and Titman (1993) in US market and by Rowenhorst (1998) in European countries, but higher than that reported by Rowenhorst (1999) in emerging markets. This difference is due to the difference in forming the portfolios. Jegadeesh and Titman (1993) and Rowenhorst (1998) define the top and bottom deciles of portfolios as winner and loser, which capture more extreme performers and generate higher winner-loser profits. While Rowenhorst (1999) refer to the top and bottom 30% stocks as winners and losers and our cutting points are top and bottom 20%. This difference makes our results not directly comparable with theirs in magnitude and significance. By using a larger sample data, and defining more extreme performers (top and bottom deciles for instance) as winners and losers as in Jegadeesh and Titman, we expect to see a higher and more significant momentum profits than documented in our study.

We also found that this pattern in stock returns is not limited to a particular stock market, but is present in eleven of sixteen emerging markets, being significant at 90% level in five of them, and marginally significant in two. For the five countries with negative momentum profits, none of the profit is significantly different from zero. What's more, winner portfolios outperform loser portfolios by more than 1% in six markets. The highest profit comes from Mexico, where winner portfolio outperform loser portfolio by about 2.21% per month, and the lowest from Brazil with monthly excess return of 0.04%.

To investigate the variation of momentum effects over time, we implement the momentum strategies for each market for two subperiods. We find that during the first sample period of 1990-1995, eleven countries show positive momentum profits and five of them are significant. On average, winners outperform losers by 0.49% per month. During the second period of 1996-2002, twelve countries have positive momentum profits and seven of them are significant and the average monthly profit across countries is 0.67%, which is higher than that during the first subperiod. Untabulated results show that from the first subperiod to the second one, the average investibility of all our sample observations increases from 35.82% to 52.39% (while the standard deviation of investibility decreases from 42.30% to 39.94%), implying that emerging markets are getting more open over time. Also we find that among the 16 markets, the momentum profits are higher in the second period than in the first one for eight markets, suggesting that momentum effect gets more pronounced in emerging markets over time. This is consistent with our conjecture that foreign investment will further drive the momentum effect: with emerging markets getting more open over time, more foreign investors are

participating in emerging markets during the latter subperiod, and thus resulting in higher momentum profits.

Aside from looking at momentum effects on individual markets, we also conduct a simple momentum strategy on stocks from all the countries with the results showed in Panel A of Table 3. The average momentum profit turns out to be 1.17% per month, which is higher than the 1% monthly return reported for developed countries (even without taking into account the difference in defining winner and loser portfolios). This is consistent with the evidence in Harvey (1995), Bekaert, Erb, Harvey and Vishanta (1997) and Chan, Hameed and Tong (2000) who document that stock returns on emerging markets are more autocorrelated and predictable. We also look at the momentum profits in subperiods and found that during the 1996-2002 period, our momentum profit was even as high as 1.93%. The increasing momentum in price over time again supports our hypothesis that participation of foreign investors drives return continuation.

2.2.2 Country-Neutral Simple Momentum Strategies

Table 2 shows that although price continuation is not constrained to particular countries, it is not present in every country. Therefore, we must construct a country-neutral momentum strategy to see how much of the momentum profit comes from country-specific factors while how much comes from firm-specific factors. Country-neutral strategy is constructed by ranking stocks from the same markets and then assigning the top (bottom) 20% of stocks with best (worst) past performance from each country as winner (loser) portfolios. Table 3 shows that the country-neutral momentum strategy generates a profit of 0.37% for the whole sample period, which is approximately

one third of the profit from the unconditional momentum strategy. It implies that around 32% of the momentum profit of 1.17% comes from firm-specific factors, while more than 60% comes from country-specific momentum effects—suggesting that not all the countries have price momentum, which is consistent with our findings in Table 2. We also notice that, by conducting country-neutral strategy, the standard deviation of momentum profits decreases from 3.48% to 1.44%, suggesting that most variance of the momentum profits is country-specific and can be diversified internationally.

2.2.3 Size-Neutral Simple Momentum Strategies

To control for size effect, we also conduct the size-neutral simple momentum strategy as follows: Firstly, at the end of each ranking period, we sort all stocks according to their size as measured by logarithm of market value in US dollars and assign each stock into one of the small, medium or large size groups. Then within each size group, stocks are ranked based on past returns and assigned to one of the five portfolios P1-P5. The winner portfolio consists of P5 stocks from each size group, while loser portfolio consists of P1 stocks from each size group. Then we compute the average monthly return of each portfolio in the following test periods. Panel B of Table 3 shows that after controlling for size, winner portfolio significantly outperform loser portfolio by 1.09% per month. Moreover, return continuation exists in all size groups and is not limited to small stocks. For size to explain the momentum profits, we expect to see a higher momentum on small stocks. However, there is a positive relation between firm size and the excess return of the momentum strategies—the profit is 0.89% for the small-

sized group while 1.33% for the large-sized groups. Therefore, we conclude that size, as a factor of risk, cannot explain the momentum profits documented in emerging markets.

2.2.4 Turnover-Neutral Simple Momentum Strategies

Several recent papers indicate that trading volume (usually measured by turnover ratio) may influence the return continuation. Lee and Swaminathan (1998) shows that both the magnitude and the persistence of price momentum can be predicted based on trading volume. Chan, Hameed and Tong (1998) document higher momentum profits on countries with higher lagged trading volume. Both papers suggest that higher trading volume may accentuate return continuation.

We examine the role of turnover in our study, defining turnover as the ratio of average monthly trading volume in the past 6 months divided by total number of shares outstanding. We conduct a turnover-neutral momentum strategy in the same way as we conduct the size-neutral strategy and report our findings in Panel C of Table 3. The main findings are: Firstly, we observe that after controlling for turnover effect, our momentum profit increases from 1.17% to 1.20% and becomes even more significant. Secondly, price continuation exists in all the three groups and all momentum profits are statistically significant. Thirdly, we didn't find the positive relationship between trading volume and momentum profits as documented in previous studies. Finally, the F statistics show that there is not any significant difference in momentum profits among the three turnover groups. Therefore, trading volume does not explain this price continuation anomaly in our study.

For all our momentum strategies documented in this section, we checked the time-variation effect by looking at the subperiod results. Again we observe that for all the 10 momentum strategies reported in Table 3, the profits from the 1996-2002 periods are higher than those from the 1990-1995 subperiod.

2.2.5 Adjustment for World Market Risk Factor

As a further test, Table 4 presents the risk-adjusted returns of all the momentum strategies implemented above. The risk-adjusted returns (reported as CAPM- α) are estimated as follows: Firstly, for each momentum portfolio, at the end of each ranking period, a monthly $\hat{\beta}$ is estimated by regressing the portfolio's past 36 months' returns in excess of the monthly risk free returns (calculated as the monthly returns of the 30-day U.S Treasury Bill) on the excess return on the Morgan Stanley Capital International All Country World Index (MSCI ACWI)⁸,

$$R_{i,t} - r_{f,t} = \alpha + \beta(R_{m,t} - r_{f,t}) + \varepsilon_{i,t} \quad t = -36, -35, \dots, -1$$

where $R_{i,t}$ is the return on the momentum portfolio (P5-P1), $r_{f,t}$ is the interest rate on 30-day Treasury Bill, and $R_{m,t}$ is the return on ACWI. Then the risk-adjusted returns (reported as CAPM- α) are calculated as:

$$\hat{\alpha}_t = (R_{i,t} - r_{f,t}) - \hat{\beta}(R_{m,t} - r_{f,t})$$

⁸ The MSCI ACWI (All Country World Index) Index is a free float-adjusted market capitalization index that is designed to measure equity market performance in the global developed and emerging markets. As of December 2003 the MSCI ACWI consisted of 49 developed and emerging market country indices.

The results in Table 4 show that the CAPM- α for the momentum portfolio is about the same as the raw returns, only becomes a little bit less significant⁹. It implies that for all the momentum strategies, whether unconditional, or country, size or turnover neutral, or within all the size and turnover subgroups, both winners and losers have about the same β s. In other words, the winner stocks do not have higher market risk than losers, and the momentum profits are not just risk premium.

3. Foreign Ownership Restriction and Momentum

Our primary objective of this paper is to find the relation between foreign ownership restriction and momentum. We use Investibility Weight Factor (IWF) as a proxy for the dispersion of ownership and the proportion of foreign investors. In this part, we conduct an IWF-based momentum strategy as follows: Firstly, at the end of each month, all stocks are ranked into one of three groups (denoted as INV1, INV2 and INV3) based on their IWF. INV1 includes stocks whose IWF is equal to 0%, which are totally closed stocks or non-investible stocks; INV3 includes stocks with IWF equal to 100%, which can be seen as highly investible stocks; and all the rest of stocks belong to INV2—partially investible stocks. Secondly, within each group, we construct the simple 6/6 month momentum strategies as we do in the previous part, then document the returns on each portfolio in test periods and calculate the zero-net-investment profits for each group.

Table 5 presents the results for our IWF-based momentum strategies. Consistent with our hypothesis, momentum profits significantly increases with IWF. The F-statistics

⁹ The lower significance level may be due to the smaller sample size in our calculating the mean of CAPM- α , since we are using rolling estimation of β (based on previous 36 months' data) in finding the risk-adjusted returns, and this method eats up the first 3 years' data. This is also the reason why we do not report the adjusted return for the subperiod of 1990-1995.

reject at 95% level the null hypothesis of equality in momentum profits between INV1 and INV3 groups. Comparing the two extreme groups, INV1 and INV3, we find something interesting: Firstly, for stocks with absolute restriction to foreign investors, as those in INV1 group, winners outperform losers by only 0.84% in return per month (with a t-statistic of 2.57). While for stocks 100% open to foreign investors, i.e., stocks in INV3 group, the monthly momentum profit can be as high as 1.98% (with a t-statistic of 2.92). This finding is consistent with our hypothesis based on Hong and Stein (1999) model: the more foreign investors holding shares, the lower the rate of information diffusion, and thus the higher the momentum profits. Secondly, we found that in INV1 group, momentum profit is heavily affected by loser stocks in P1. In fact, buying winner stocks even attenuates the profitability of this trading strategy. By just shorting losers, investors can make a higher profit of 1.03%. This is also consistent with Hong, Lim and Stein (2000) who find that bad news diffuses more slowly among investors than good news. But if we look at the momentum profit for INV3 group, we found that momentum profit comes from both winner and loser sides. In particular, we found that while loser portfolio of INV3 group even underperforms loser portfolio in INV1 group, winner portfolio of INV3 group outperforms that of INV1 group. This evidence shows that when the number of foreign investors increases, investors underreact not only to bad news, but also to good news. This also makes intuitive economic sense. For closed stocks as in INV1 group, because managers have incentive to push good news out to investors as fast as possible, but to delay the disclosure of bad news as late as possible, the slow diffusion of bad news causes investors to underreact to bad news, thus create momentum effect in losers. But for highly open stocks like those in INV3 group, quite a proportion of the investors are

foreigners. Because of the cultural, linguistic as well as geographic barriers we discussed before, even after managers announce the good news publicly, foreign investors, relative to domestic investors, are still slow in accessing, obtaining and digesting these information. The higher the proportion of foreign investors, i.e., the higher the dispersion of shareholders, then the more slowly the information diffuses among them, even it is good news. Therefore, both winner and loser stocks in INV3 group show return continuation.

The subperiods tests also show the same pattern—the highly open stocks have stronger continuation than the highly closed stocks. And the difference is becoming more significant over time.

Though the results are quite consistent with Hong and Stein (1999)'s slow information diffusion model, there might be other alternative explanations to this phenomenon. Next, we are going to investigate other possible explanations.

3.1. Simple Risk Characteristics of IWF-Based Portfolios

One possible explanation for momentum profit is that the excess return is just compensation for risk. Bae, Chan and Ng (2004) find that highly investible emerging market stocks have higher world market risks than the non-investible stocks. Chari and Henry (2001) also show that there is a difference in the world market β s between investible and non-investible stocks. It is possible that the higher returns on highly investible stocks are just risk premium for their larger exposure to world market risks. To test this hypothesis, we compute the β s for each portfolio as we constructed in Table 5. Again, we use MSCI All Country World Index (ACWI) to calculate the world market

returns and the β s of each portfolio as are reported in Table 6. Consistent with other researches, we also find that the highly investible stocks are subject to greater world market exposure than the non-investible stocks: the mean β for totally open stocks is 1.36 while that for closed stocks is only 0.69¹⁰. Therefore, we conjecture that at least part of the higher momentum returns from highly open stocks are compensation for their larger exposure to world market risk. Later we will try to find out how much of the profits come from the systematic risk compensation.

However, if we compare the β s of winner and loser portfolios within the same INV groups, we may find that momentum profits on each INV group seem unlikely to be explained by their covariance with the world markets. For market risk to explain the excess return, the β of winner portfolio should be higher than that of loser portfolios. However, from Table 6 we find that, in all the three INV groups, the β s of the losers (P1) are even higher than those of the winners (P5). Therefore, the momentum profits documented within each INV group is not because of world market risk, but related to country or firm specific factors.

To test whether size can explain our momentum profits, we also report the size for each portfolio in Table 6. We have two main findings: Firstly, for size to explain the excess return of winner portfolios, it should be larger for losers than for winners. Table 6 shows that, in all INV groups, winners are larger in market capitalization than losers. For example, in INV1 group, the size of loser portfolio is 3.63, while that of winner portfolio is 4.33 and similar case shows up in the other two INV groups. Overall, the average size of losers is 4.59, while that of winners is 5.57. Therefore, size as a factor of risk, cannot

¹⁰ We also used EMDB Composite index to calculate the β s of all the portfolios and get the similar results, with the mean β s of 1.02 for the highly investible stocks and 0.60 for the non-investible stocks.

explain the momentum profits within each INV group. Secondly, size cannot explain the higher excess return of highly investible stocks than non-investible stocks either for the same reason. On average, the size of highly investible stocks in INV3 group is 5.64 while that of non-investible stocks is 4.01. Also we will investigate the effect of size on the INV-Based momentum strategies later.

The next 5 columns of Table 6 report the average turnover ratio for each portfolio. Again, for trading volume to explain the excess return of winner portfolios, it should be larger for winners than for losers. We find that for the first and second INV groups, turnover ratios of winners are lower than those of losers. For the third INV group, the turnover ratio of winners is only slightly higher than that of losers. Therefore, the momentum returns on each INV group of stocks cannot be driven by trading volume. However, comparing the average turnover ratio among the three INV categories, we find that the turnover ratio of the highly investible stocks (0.1539) is significantly higher than that of the non-investible stocks (0.0784). Thus we cannot rule out the possibility that the higher momentum returns from highly investible stocks are driven by the higher trading volume.

The coarse descriptive statistics of risk characteristics of IWF-based portfolios only give us a rough idea of how other risk factors like firm size can affect the profitability of our IWF-based momentum strategies. For our results to be robust, we need more specific tests.

3.2. Size-Neutral, IWF-Based Momentum Strategies

The investability measurement—IWF, may capture some other factors of the firm such as size. For example, companies with greater foreign ownership restrictions tend to be smaller firms while large firms are more likely to be open to foreign investors and are more likely to be attractive to them. Therefore, in this section we explore the effectiveness of IWF-based momentum strategies that avoid taking significant size positions, so as to separate the influence of size effect.

To do so, we construct the size-neutral IWF-based strategies as follows: at the end of each ranking period, we first categorize stocks based on their investability and assign each stock to INV1, INV2 and INV3 groups as before. Then within each INV group, stocks are again sorted by their size and assigned to each of small, medium and large size categories. This gives us 9 double sorted groups. Within each of the size-INV groups, we rank stocks by their ranking period returns and assign them to one of P1-P5 five portfolios, with P1 as losers and P5 as winners. Then we can construct our size-neutral IWF-based strategies: for each INV category, the winner portfolio consists of winner stocks (P5) stocks from each of the three size groups, while loser portfolio consists of loser stocks (P1) stocks also from each size group. And the momentum profits are the difference in returns between winner and loser portfolios. Table 7 Panel A reports the size-neutral IWF-based momentum returns. We find that, after controlling for size, the momentum profits in each IWF categories do not differ much from those in Table 5.

Consistent with our findings in Table 6, size does not explain the excess momentum returns from highly investible stocks. The F-statistics for testing the equality in momentum returns between INV1 and INV3 groups becomes even more significant.

Like in previous parts, we also conduct the subperiod tests. For both subperiods, the momentum strategies in INV3 generate higher returns than those in INV1. The difference is getting larger and more significant over time: the momentum strategy on highly investible stocks dominates that on non-investible stock by 0.55% per month in the first subperiod and by 1.83% in the second subperiod.

3.3. Turnover-Neutral, IWF-Based Momentum Strategies

We document in Table 6 that the average turnover ratio of highly investible stocks is higher than that of non-investible stocks. Therefore it is possible that participation of foreign investors increases the trading volume of the stocks, thus drives the price continuation. Therefore we explore the turnover-neutral IWF-based momentum strategies to separate the influence of trading volume. Using similar methods as in previous section, we construct the turnover-IWF double sorted momentum strategies. Table 7 Panel B reports the momentum profits within each of the turnover-neutral INV groups and in subperiods. We find that the adjustment for trading volume almost does not change the momentum profits presented in Table 5. Excess return on highly open stocks still dominates that on closed stocks. And the difference is getting more significant over time.

3.4. Country-Neutral, IWF-Based Momentum Strategies

In the previous part, we find that part of the momentum profits come from specific countries—controlling for country, the momentum profits decrease by nearly two thirds. Therefore, we want to see whether our INV effect (highly investible stocks have

higher momentum than non-investible stocks) is also country-specific by conducting IWF-based momentum strategies conditional on countries.

Table 7 panel C reports the country-neutral IWF-based momentum profits. After controlling for country, the momentum profits of all the strategies decrease. For the whole sample period, the profit for INV1 group decreases from 0.84% (with t-statistic of 2.57) to 0.18% (with t-statistic of 0.75), and that for INV3 group decreases from 1.98% (with t-statistic of 2.92) to 0.82% (with t-statistic of 2.91). The significant decrease in excess returns implies that the INV effect may not be pervasive, or may differs in extent, among all the sample countries. However, looking at the F-statistics, we find that controlling for country still cannot fully eliminate the return difference between highly investible stocks and non-investible stocks. In fact, the difference becomes even more significant for the whole sample period as well as for the first subperiod.

To find out exactly which countries have the INV effect, we implement the IWF-based strategies in each of the 16 sample markets. Table 8 shows that the INV effect is absent on only 5 markets—Chile, Malaysia, Taiwan of China, Turkey and Venezuela. For Jordan and Thailand, we cannot test the existence of INV effect due to the small sample size. But we do find that the partially open stocks (INV2) have stronger continuation effect than the totally non-investible stocks (INV1). For the rest 9 markets, the totally investible stocks all generate higher momentum returns than the non-investible stocks, though as we expected, with different extent: The return difference ranges from 2.73% per month in Indonesia to 0.39% in Greece. The average return difference of the 16 markets is 0.19% per month.

3.5. Market Risk Adjustment for IWF-Based Momentum Strategies

Recall that in Table 6, we document a higher average β for highly investible stocks than for non-investible stocks, which implies that at least part of the higher momentum returns from highly open stocks are compensation for their larger exposure to world market risk. Thus we are interested in finding out how much of the profits come from the systematic risk compensation by adjusting the world market risk for all the IWF-based momentum returns.

Table 9 Panel A reports the CAPM- α of the momentum returns within each INV group. After the adjustment for world market risk, the momentum return in non-investible stocks decreases from 0.84% to 0.64%, while that in highly investible stocks decreases even more—from 1.98% to 1.58%. The return difference between these two INV groups decreases from 1.14% in raw return to 0.94% in CAPM- α . These results imply that about one fifth of the higher momentum profits in highly investible stocks come from the compensation for their higher exposure to the world market systematic risks, while the rest may come from country or firm specific factors.

To further decompose the return difference, we report the country-neutral IWF-based momentum returns as well as their market adjusted CAPM- α in Table 9 Panel B. It shows that the removal of country specific factors decreases the return difference between INV1 and INV3 from 1.14% to 0.66%, and the adjustment for world market risk further decreases the return difference to 0.50%, which is still significant at 95% level. We can see that, about half of the excess momentum returns from the highly investible stocks (less that from the non-investible stocks) can be explained by world market risk or country specific factors, while the other half remains unexplained.

3.6. Foreign Investors as Momentum Traders?

Another possible explanation is that high momentum profits among stocks with high foreign ownership could be caused by the positive feedback trading of foreign investors. Studies on foreign investors' trading behaviour show that when foreign investors have disadvantage in information, they tend to trade on past returns. For instance, Williamson (1993) suggests that foreigners may tend to show herdlike behaviour in their investment decision. Richards (1996) attributes the short-horizon positive autocorrelation of return in emerging markets to the herding behaviour of a group of uninformed traders, including foreign investors. Brennan and Cao (1997) examine U.S. portfolio investment in emerging markets and discern that U.S. purchases are positively associated with local current and one-quarter-lagged market returns in many countries. More evidence on international investors' positive feedback trading comes from Tesar and Werner (1995a, 1995b), Bohn and Tesar (1996), and Froot, O'Connell and Seasholes (2001). If foreign investors trade on past returns, their trading will also cause positive autocorrelation, or momentum, in stock returns. Without transaction data, it is hard to clearly separate the momentum effects caused by foreign investors' positive feedback trading behaviour from those caused by slow information diffusion among investors. However, positive feedback trading will only cause an underreaction in stock returns, while the slow information diffusion model predicts not only an initial underreaction but also a reversal in stock returns at a longer horizon. Therefore, looking at the persistence of momentum profitability by implementing the momentum strategies at longer horizons can help us at least partially differentiate the two explanations.

Table 10 presents the momentum profits where ranking and test periods are 6, 12, 24 and 36 months separately. We also report the returns at different horizons after controlling for country, size and turnover ratio. Consistent with Richards (1996), we find that the positive autocorrelation turns to be negative at horizons of one year or more with economic significance (though some are not statistically significant). For example, at one year horizon, the prior loser portfolio outperforms prior winner portfolios by 3.65% per month (significant at 90% level) for the unconditional momentum strategy, or by 0.47%, 3.71% and 3.44% per month after controlling for country, size and turnover separately.

The fact that the short-run momentum returns reverse in emerging markets is consistent with the gradual-information-diffusion model, which states that “the very existence of underreaction sows the seeds for overreaction (Hong and Stein, 1999)”.

Table 11 shows our IWF-based momentum returns at different horizons. Similarly, for all the three INV groups, the momentum profits turn to be negative at horizons of one year or more (except for partially investible stocks at the 2 year horizon). And in Panel B of Table 11 we find that after controlling for country, the subsequent reversal for non-investible stocks disappears, but that for highly investible stock is still there, though it takes place at a longer horizon of 2 years. This is also consistent with our conjecture that the higher foreign ownership drives the short-run underreaction and the subsequent reversal.

Therefore, the results are more consistent with the prediction by gradual information diffusion model. Though we cannot totally rule out the possibility of momentum trading by foreign investors, we can be confident that momentum trading alone cannot drive the results we find.

3.7. Foreign investors Overreact?

Daniel, Hirshleifer, and Subrahmanyam (1998) present a model where investors are assumed to be psychologically biased: they are overconfident about their private information and have self-attribution bias when they receive confirming or disconfirming information. Such biases cause investors to overreact to their private information and generate return momentum. The overreaction in prices will eventually get corrected in the long run. Thus, Daniel et.al (1998)'s overconfidence model also predicts an initial continuation and a later reversal in return. Therefore, another possibility for our empirical finding could be that, foreign investors are generally more overconfident about and thus more likely to overreact to their private information.

We investigate this explanation by looking at the market model regression results for stocks with different level of foreign ownership. The logic behind is that, if foreign investors are on average more overconfident than domestic investors and they are more likely to overreact to their private news, stocks with high foreign ownership will tend to deviate from fundamental values more frequently than stocks with low foreign ownership, *ceteris paribus*. That means, the price of stocks with high foreign ownership will be less efficient, and the market model will generate a lower R^2 on average. In empirical test, we limit our sample stocks to those that are consistently totally open or close to foreign investors, that is, stocks that have $INV=1$ or $INV=0$ for our whole sample period. Then each year, we run the market model regression for each stock, and calculate the average R^2 across stocks within the same INV level. To increase the power of the test, we use weekly returns and extend the sample period to the earliest date that the data is available

for individual stock from the database. Test results are reported in Table 12. For overreaction to explain our empirical finding, the average R^2 should be higher for the non-investible stocks (INV=0) than for the highly investible stocks (INV=1). However, Table 12 shows that the average R^2 is 29.52% for the highly investible stocks and 19.05% for the non-investible stocks—a result that is contrary to the overreaction hypothesis. As a robustness test, we change our cutting points and define highly investible stocks as stocks with average INV over the sample period greater than or equal to 0.8, and non-investible stocks as those with average INV less than or equal to 0.2. We rerun the test and get the same results as shown in Table 12. The highly investible stocks have an average R^2 of 34.13% while the non-investible stocks' average R^2 is 19.55%.

We also conduct other robustness check by adding more lead and lag market returns in the market model, or regressing R^2 from market model for each individual stock on its INV. All the results are very robust¹¹. Therefore, the results do not support the overreaction hypothesis.

4. Conclusion

Since the mid-1980s, the majority of emerging markets have started liberalizing their equity markets to international investors. As one of the studies on the impact of liberalization on the economies of these markets, this paper examines the impact of foreign ownership restriction on the predictability of stock returns, or in particular, on the profitability of momentum strategies. Foreign investors, compared with domestic investors, are at information disadvantage due to the cultural, linguistic and sometimes

¹¹ Tabulated results are available upon request.

geographic barriers. Therefore, their participation in the emerging markets will inevitably affect the return patterns. In particular, we expect to see that for stocks that are more open to foreign investors, or have higher foreign ownership, as information diffuses less efficiently among investors, there should be higher momentum in returns.

Our results confirm this conjecture. We find that, for stocks with different accessibility to foreign investors, the profitability of momentum strategies is significantly different. In particular, the momentum strategy on highly investible stocks significantly outperforms the strategies on non-investible stocks. We also find that this outperformance is not totally because of market risk, size, turnover or country-specific factors, and cannot be totally explained by foreign investors' positive feedback trading, or their overreaction to private information. It is more consistent with Hong and Stein (1999)'s gradual information diffusion theory. And consistent with their model's underreaction-overreaction prediction, short term return continuation finally reverses in a longer period of time.

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Table 1 Descriptive Statistics of Sample Countries

This table presents the descriptive statistics for the sample markets. The first two columns of data are the number of observations and number of stocks sampled in each market. The next six columns report the descriptive statistics for Investibility Weight Factor, a degree-open factor, representing the degree to which foreign investors may access the stock. The next 2 columns report the average monthly returns in US dollars for these 16 markets for the whole sample. To investigate the variation of stock markets over time, we divide our sample period into two sub-periods: January 1990 – December 1995 and January 1996 – July 2002. The size reported in the last three columns of Table 1 is measured as the natural logarithm of the market value of sample stocks (in US dollars).

Country	No. of Obsers.	No. of Stoc ks	IWF							Monthly Returns						Size		
			Min	Max	25%	75%	Mean	Media n	Std. Dev	Mean (90-02)	Std. Dev	Mean (90-95)	Std. Dev	Mean (96-02)	Std. Dev	Mea n	Medi an	Std. Dev
Argentina	4008	46	0.00	1.00	0.76	1.00	0.77	1.00	0.40	0.0121	0.2971	0.0380	0.3942	-0.0114	0.1615	5.31	5.28	1.61
Brazil	10758	135	0.00	1.00	0.00	1.00	0.59	0.75	0.42	0.0172	0.2560	0.0336	0.3202	0.0047	0.1927	5.77	5.75	1.72
Chile	5796	58	0.00	1.00	0.07	1.00	0.50	0.25	0.42	0.0067	0.1254	0.0261	0.1371	-0.0070	0.1145	6.09	6.03	1.17
Colombia	3078	22	0.00	1.00	0.00	1.00	0.42	0.00	0.46	0.0070	0.1700	0.0242	0.1851	-0.0077	0.1544	4.98	5.17	1.38
Greece	6185	95	0.00	1.00	0.70	1.00	0.79	1.00	0.37	0.0082	0.1606	0.0040	0.1416	0.0105	0.1701	5.28	5.21	1.42
Indonesia	6872	78	0.00	1.00	0.00	0.49	0.45	0.49	0.36	0.0058	0.2653	-0.0008	0.1389	0.0117	0.3400	4.72	4.73	1.69
Jordan	4990	66	0.00	1.00	0.00	0.00	0.09	0.00	0.21	-0.0032	0.0874	0.0002	0.0856	-0.0055	0.0886	3.17	3.11	1.43
Korea	18689	207	0.00	1.00	0.10	0.81	0.39	0.20	0.37	0.0061	0.2423	-0.0006	0.1122	0.0104	0.2967	5.52	5.50	1.41
Malaysia	14420	121	0.00	1.00	0.78	1.00	0.83	1.00	0.29	0.0050	0.2169	0.0144	0.1339	0.0001	0.2492	5.83	5.96	1.45
Mexico	8758	45	0.00	1.00	0.73	1.00	0.77	1.00	0.41	0.0071	0.1601	0.0067	0.1548	0.0074	0.1649	6.07	6.11	1.49
Pakistan	7518	63	0.00	1.00	0.00	0.70	0.26	0.00	0.43	0.0052	0.1629	0.0109	0.1351	-0.0010	0.1882	3.60	3.52	1.34
Philippines	5907	45	0.00	1.00	0.00	0.40	0.32	0.30	0.37	-0.0014	0.2239	0.0097	0.1575	-0.0079	0.2544	5.02	4.88	1.72
Taiwan	12179	152	0.00	1.00	0.03	0.50	0.24	0.20	0.19	-0.0001	0.1676	0.0026	0.1558	-0.0021	0.1754	6.53	6.40	1.31
Thailand	8039	120	0.00	1.00	0.20	0.40	0.29	0.25	0.17	-0.0005	0.2491	0.0099	0.1472	-0.0079	0.3004	5.63	5.65	1.44
Turkey	5859	85	0.00	1.00	0.62	1.00	0.81	1.00	0.31	0.0194	0.2517	0.0238	0.2607	0.0172	0.2470	5.38	5.24	1.46
Venezuela	2210	20	0.00	1.00	0.00	1.00	0.55	1.00	0.49	0.0119	0.2084	0.0261	0.2222	-0.0009	0.1944	4.89	5.05	1.39
Mean			0.00	1.00	0.25	0.77	0.50	0.53	0.35	0.0067	0.2028	0.0143	0.1801	0.0007	0.2058	5.24	5.22	1.46

Table 2
Monthly Returns to Simple Momentum Strategies on Individual Countries

This table presents the profitability of simple momentum strategies in the 16 emerging markets from January 1990 to July 2002. The 6/6 month momentum portfolios constructed are similar to those in Jegadeesh and Titman (1993): at the end of each month, all sample stocks are sorted based on their past 6-month (ranking period) returns and then assigned to one of five quintiles (P1 represents the worst past performers—defined as losers, and P5 represents the best past performers—defined as winners). The portfolios are equally weighted and held for subsequent 6 months (test period). The performance of these five portfolios in the subsequent 6 months test period are calculated. We define the return to a momentum (or zero-net-investment) strategy as the return on the winner portfolio (P5) less the return on the loser portfolio (P1). To investigate the variation of momentum effects over time, we also implement the momentum strategies for each market for two subperiods: January, 1990—December, 1995 and January, 1996—July, 2002. The t-statistics reported are computed using the Newey-West correction for heteroskedasticity and autocorrelation.

	Test Period Returns (1990-2002)					1990-1995	1996-2002	
	P1	P2	P3	P4	P5	P5-P1	P5-P1	P5-P1
Argentina	-0.0072	-0.0096	-0.0038	-0.0005	0.0060	0.0131	0.0005	0.0256
<i>t-statistic</i>	-0.47	-0.91	-0.33	-0.05	0.56	1.62	0.03	3.12
Brazil	-0.0007	-0.0019	-0.0015	0.0006	-0.0004	0.0004	-0.0099	0.0105
<i>t-statistic</i>	-0.08	-0.23	-0.20	0.09	-0.04	0.06	-1.01	1.65
Chile	-0.0044	0.0020	0.0052	0.0066	0.0072	0.0116	0.0148	0.0083
<i>t-statistic</i>	-0.84	0.37	0.92	1.14	1.12	3.16	2.68	1.80
Colombia	-0.0069	-0.0062	0.0011	0.0077	0.0052	0.0121	0.0160	0.0083
<i>t-statistic</i>	-0.77	-0.84	0.13	0.84	0.66	1.73	1.91	0.74
Greece	-0.0101	-0.0041	-0.0033	-0.0014	0.0035	0.0136	0.0113	0.0159
<i>t-statistic</i>	-1.35	-0.47	-0.39	-0.16	0.34	2.58	1.69	1.96
Indonesia	-0.0156	-0.0208	-0.0199	-0.0177	-0.0143	0.0013	0.0019	0.0007
<i>t-statistic</i>	-1.10	-1.56	-1.52	-1.33	-1.12	0.25	0.33	0.08
Jordan	-0.0062	-0.0054	-0.0042	0.0002	0.0003	0.0065	0.0023	0.0107
<i>t-statistic</i>	-1.46	-1.48	-1.19	0.05	0.08	1.61	0.48	1.67
Korea	-0.0154	-0.0096	-0.0097	-0.0102	-0.0167	-0.0013	-0.0061	0.0035
<i>t-statistic</i>	-1.42	-0.86	-0.91	-0.99	-1.80	-0.31	-1.64	0.47
Malaysia	-0.0064	-0.0063	-0.0051	-0.0049	-0.0088	-0.0024	-0.0017	-0.0031
<i>t-statistic</i>	-0.61	-0.64	-0.51	-0.54	-0.96	-0.57	-0.51	-0.40
Mexico	-0.0137	-0.0051	-0.0006	0.0036	0.0085	0.0221	0.0288	0.0155
<i>t-statistic</i>	-1.78	-0.71	-0.09	0.51	1.21	5.39	7.17	2.30
Pakistan	-0.0050	-0.0010	-0.0045	-0.0018	-0.0052	-0.0001	0.0082	-0.0083
<i>t-statistic</i>	-0.63	-0.14	-0.58	-0.24	-0.67	-0.04	2.27	-1.34
Philippines	-0.0179	-0.0139	-0.0071	-0.0107	-0.0119	0.0060	0.0131	-0.0010
<i>t-statistic</i>	-1.96	-1.63	-0.83	-1.30	-1.39	1.05	1.40	-0.16
Taiwan	-0.0079	-0.0094	-0.0076	-0.0111	-0.0132	-0.0053	-0.0123	0.0016
<i>t-statistic</i>	-0.87	-1.19	-1.01	-1.47	-1.53	-1.07	-2.08	0.21
Thailand	-0.0204	-0.0165	-0.0149	-0.0119	-0.0068	0.0136	0.0107	0.0164
<i>t-statistic</i>	-1.76	-1.54	-1.47	-1.21	-0.81	2.39	1.60	1.81
Turkey	-0.0082	-0.0050	-0.0052	-0.0023	-0.0083	-0.0002	0.0034	-0.0037
<i>t-statistic</i>	-0.69	-0.42	-0.45	-0.19	-0.74	-0.03	0.29	-0.62
Venezuela	-0.0031	-0.0036	-0.0082	0.0022	-0.0015	0.0016	-0.0033	0.0064
<i>t-statistic</i>	-0.24	-0.30	-0.71	0.17	-0.13	0.22	-0.32	0.73
MEAN	-0.0093	-0.0073	-0.0056	-0.0032	-0.0035	0.0058	0.0049	0.0067

Table 3
Returns of Country-, Size- and Turnover-Based Momentum Strategies

Panel A reports the simple momentum profits on all stocks from the 16 countries, as well as those on a country-neutral momentum strategies to see how much of the profits comes from country-specific factors. Country-neutral strategy is constructed by ranking stocks from the same markets and then assigning the top (bottom) 20% of stocks with best (worst) past performance from each country as winner (loser) portfolios. Panel B reports the profitability of size-neutral momentum strategies: Firstly, at the end of each ranking period, all stocks are sorted according to their size as measured by logarithm of market value in US dollars and assigned into one of the small, medium or large size groups. Then within each size group, stocks are ranked based on past returns and assigned to one of the five portfolios P1-P5. The winner portfolio consists of P5 stocks from each size group, while loser portfolio consists of P1 stocks from each size group. Then average monthly return of each portfolio in the following test periods are calculated and presented. Panel B also reports the momentum profits within each size groups. Panel C reports the profitability of turnover-neutral strategies which are constructed in the same way as size-neutral strategies, where turnover is defined the ratio of average monthly trading volume in the past 6 months divided by total number of shares outstanding. We also present the momentum profits within each turnover groups. As before, for each momentum strategy in each panel, subperiods results are also reported. The t-statistics reported are computed using the Newey-West correction for heteroskedasticity and autocorrelation.

Test Period Returns (1990-2001)						1990-1995	1996-2001
P1	P2	P3	P4	P5	P5-P1	P5-P1	P5-P1
A: Returns of Country-Based Momentum Strategies							
All Stocks							
Mean	-0.0117	-0.0122	-0.0082	-0.0023	0.0000	0.0117	0.0039
t-statistic	-1.47	-1.92	-1.48	-0.44	0.01	2.34	0.93
Country-neutral							
Mean	-0.0098	-0.0079	-0.0055	-0.0050	-0.0061	0.0037	0.0024
t-statistic	-1.56	-1.33	-0.94	-0.90	-1.10	1.76	0.98
B: Returns of Size-Based Momentum Strategies							
Size-neutral							
Mean	-0.0113	-0.0118	-0.0079	-0.0027	-0.0004	0.0109	0.0031
t-statistic	-1.47	-1.83	-1.43	-0.52	-0.07	2.30	0.73
Small							
Mean	-0.0126	-0.0148	-0.0108	-0.0052	-0.0037	0.0089	0.0050
t-statistic	-1.51	-2.11	-1.76	-0.88	-0.68	1.67	0.98
Medium							
Mean	-0.0116	-0.0126	-0.0098	-0.0032	-0.0005	0.0112	0.0026
t-statistic	-1.48	-1.99	-1.70	-0.63	-0.09	2.22	0.62
Large							
Mean	-0.0101	-0.0086	-0.0034	0.0002	0.0033	0.0133	0.0033
t-statistic	-1.41	-1.43	-0.69	0.03	0.60	2.83	0.60
F-statistics						0.5257	0.1344
(P-value)						0.5915	0.8743
C: Returns of Turnover-Based Momentum Strategies							
Turnover-neutral							
Mean	-0.0113	-0.0130	-0.0081	-0.0025	0.0007	0.0120	0.0045
t-statistic	-1.48	-1.99	-1.45	-0.48	0.13	2.46	1.06
Low							
Mean	-0.0111	-0.0112	-0.0069	-0.0017	0.0028	0.0138	0.0074
t-statistic	-1.65	-2.02	-1.43	-0.37	0.56	3.32	1.42
Medium							
Mean	-0.0075	-0.0115	-0.0049	-0.0003	0.0028	0.0103	0.0041
t-statistic	-0.99	-1.72	-0.83	-0.05	0.51	1.98	0.98
High							
Mean	-0.0162	-0.0165	-0.0127	-0.0057	-0.0037	0.0125	0.0030
t-statistic	-1.8	-2.22	-2.1	-1.02	-0.66	2.03	0.57
F-statistics						0.3439	0.4707
(P-value)						0.7092	0.6252

Table 4
World Market Risk-Adjusted Returns

Table 4 presents the risk-adjusted returns of all the momentum strategies implemented above. The risk-adjusted returns (reported as CAPM- α) are estimated as follows: Firstly, for each momentum portfolio, at the end of each ranking period, a monthly $\hat{\beta}$ is estimated by regressing the portfolio's past 36 months' returns in excess of the monthly risk free returns (calculated as the monthly returns of the 30-day U.S Treasury Bill) on the excess return on the Morgan Stanley Capital International All Country World Index (MSCI-ACWI),

$$R_{i,t} - r_{f,t} = \alpha + \beta(R_{m,t} - r_{f,t}) + \varepsilon_{i,t} \quad t = -36, -35, \dots, -1$$

where $R_{i,t}$ is the return on the momentum portfolio (P5-P1), $r_{f,t}$ is the interest rate on 30-day Treasury Bill, and $R_{m,t}$ is the return on ACWI. Then the risk-adjusted returns (reported as CAPM- α) are calculated as:

$$\hat{\alpha}_t = (R_{i,t} - r_{f,t}) - \hat{\beta}(R_{m,t} - r_{f,t})$$

The t-statistics reported are computed using the Newey-West correction for heteroskedasticity and autocorrelation.

		1990-2002		1996-2002	
		Raw	CAPM- α	Raw	CAPM- α
A: Returns of Country-Based Momentum Strategies					
All Stocks					
Mean		0.0117	0.0119	0.0193	0.0174
t-statistic		2.34	1.78	2.25	1.81
Country-neutral					
Mean		0.0037	0.0033	0.0050	0.0013
t-statistic		1.76	1.58	1.49	0.46
B: Returns of Size-Based Momentum Strategies					
Size-neutral					
Mean		0.0109	0.0099	0.0187	0.0159
t-statistic		2.3	2.42	2.33	2.75
Small					
Mean		0.0089	0.0064	0.0129	0.0100
t-statistic		1.67	1.49	1.38	1.61
Medium					
Mean		0.0112	0.0115	0.0198	0.0173
t-statistic		2.22	3.3	2.29	3.51
Large					
Mean		0.0133	0.0146	0.0234	0.0237
t-statistic		2.83	2.47	3.49	3.09
F-statistics		0.5257	1.5027	1.1802	1.7385
(P-value)		0.5915	0.2241	0.3092	0.1782
C: Returns of Turnover-Based Momentum Strategies					
Turnover-neutral					
Mean		0.0120	0.0127	0.0194	0.0176
t-statistic		2.46	1.96	2.33	1.90
Low					
Mean		0.0138	0.0144	0.0202	0.0183
t-statistic		3.32	2.96	3.34	2.66
Medium					
Mean		0.0103	0.0111	0.0164	0.0154
t-statistic		1.98	1.55	1.78	1.48
High					
Mean		0.0125	0.0122	0.0220	0.0193
t-statistic		2.03	1.52	2.08	1.71
F-statistics		0.3439	0.2025	0.3200	0.1305
(P-value)		0.7092	0.8168	0.7265	0.8777

Table 5
Returns to IWF-Based Momentum Strategies on All Stocks

Table 5 reports the return to IWF-based momentum strategies on all stocks from the 16 sample countries, both in the whole sample period and in the two subsample periods. The IWF-based momentum strategies are constructed as follows: Firstly, at the end of each month, all stocks are ranked into one of three groups (denoted by INV1-INV3) based on their "Foreign Investable Weight Factor (IWF)", which is a degree-open factor from the Standard & Poor Emerging Market Database (EMDB), representing the degree to which foreign investors may access the stocks. INV1 includes stocks whose IWF is equal to 0%, which are totally closed stocks or non-investible stocks; INV3 includes stocks with IWF equal to 100%, which can be seen as highly investible stocks; and all the rest of stocks belong to INV2—partially investible stocks. Secondly, within each group, we construct the simple 6/6 month momentum strategies as we do in the previous part, then document the returns on each portfolio in test periods and calculate the zero-net-investment profits for each group. The t-statistics reported are computed using the Newey-West correction for heteroskedasticity and autocorrelation.

INV Groups		Test Period Returns						1990-1995	1996-2002
		P1	P2	P3	P4	P5	P5-P1	P5-P1	P5-P1
INV1	Mean	-0.0103	-0.0118	-0.0068	-0.0025	-0.0018	0.0084	0.0078	0.0090
	<i>t</i>	-1.76	-2.89	-1.61	-0.52	-0.38	2.57	1.53	2.18
INV2	Mean	-0.0091	-0.0119	-0.0109	-0.0058	-0.0050	0.0041	-0.0097	0.0177
	<i>t</i>	-1.03	-1.71	-1.72	-0.99	-0.85	0.67	-2.12	1.73
INV3	Mean	-0.0131	-0.0093	-0.0004	0.0050	0.0067	0.0198	0.0128	0.0267
	<i>t</i>	-1.63	-1.44	-0.08	0.94	1.04	2.92	1.8	2.36
F-statistics for INV1=INV3							5.9792	0.7720	5.8715
<i>(P-value)</i>							0.0151	0.3811	0.0166

Table 6
Risk Characteristics of IWF-based Portfolios

Table 6 reports the risk characteristics of the IWF-based Portfolios. MSCI All Country World Index (ACWI) are used to calculate the world market returns and the β s of each portfolio. Size is defined as the logarithm of market value in US dollars. Trnover is defined the ratio of average monthly trading volume in the past 6 months divided by total number of shares outstanding.

	N	Mean IWF	Mean Beta	Mean Size	Mean TO.	Mean Beta of Portfolios					Mean Size					Mean Turnover				
						P1 (L)	P2	P3	P4	P5 (W)	P1 (L)	P2	P3	P4	P5 (W)	P1 (L)	P2	P3	P4	P5 (W)
INV1	39174	0.0	0.6885	4.0175	0.0784	0.8919	0.6648	0.5726	0.6160	0.6996	3.63	3.98	4.00	4.20	4.33	0.0969	0.0580	0.0562	0.0703	0.1095
INV2	60704	0.4	1.3361	6.0156	0.1419	1.4511	1.3158	1.2751	1.2743	1.3646	5.32	5.87	6.15	6.33	6.43	0.1270	0.1185	0.1240	0.1487	0.1987
INV3	39218	1.0	1.3595	5.6437	0.1539	1.4940	1.3436	1.2981	1.2698	1.3955	4.81	5.51	5.83	6.01	5.95	0.1081	0.1041	0.3524	0.1158	0.0981
Aver.		0.5124	1.1280	5.2256	0.1247	1.2790	1.1081	1.0486	1.0534	1.1532	4.59	5.12	5.33	5.51	5.57	0.1107	0.0935	0.1775	0.1116	0.1354

Table 7
Risk Adjustment for IWF-based Momentum Strategies

A: Size-Neutral IWF-Based Momentum Strategies

Table 7 Panel A reports the size-neutral IWF-based momentum returns. The size-neutral IWF-based strategies are constructed as follows: at the end of each ranking period, we first categorize stocks based on their investibility and assign each stock to INV1, INV2 and INV3 groups as before. Then within each INV group, stocks are again sorted by their size and assigned to each of small, medium and large size categories. This gives us 9 double sorted groups. Within each of the size-INV groups, we rank stocks by their ranking period returns and assign them to one of P1-P5 five portfolios, with P1 as losers and P5 as winners. Then we can construct our size-neutral IWF-based strategies: for each INV category, the winner portfolio consists of winner stocks (P5) stocks from each of the three size groups, while loser portfolio consists of loser stocks (P1) stocks also from each size group. And the momentum profits are the difference in returns between winner and loser portfolios. The t-statistics reported are computed using the Newey-West correction for heteroskedasticity and autocorrelation.

INV Groups	Test Period Returns						1990-1995	1996-2001
		P1	P2	P3	P4	P5	P5-P1	P5-P1
INV1	Mean	-0.0098	-0.0101	-0.0078	-0.0031	-0.0022	0.0077	0.0073
	<i>t</i>	-1.71	-2.34	-1.91	-0.62	-0.44	2.36	2.00
INV2	Mean	-0.0091	-0.0108	-0.0110	-0.0067	-0.0054	0.0037	-0.0102
	<i>t</i>	-1.06	-1.54	-1.70	-1.12	-0.91	0.63	1.81
INV3	Mean	-0.0129	-0.0082	-0.0007	0.0047	0.0067	0.0196	0.0128
	<i>t</i>	-1.62	-1.28	-0.12	0.86	1.05	2.98	2.42
F-statistics for INV1=INV3							6.8304	0.9424
<i>(P-value)</i>							0.0094	0.3333
								6.6729
								0.0108

B: Turnover-Neutral IWF-Based Momentum Strategies

Table 7 Panel B reports the turnover-neutral IWF-based momentum returns. The turnover-neutral IWF-based strategies are constructed as follows: at the end of each ranking period, we first categorize stocks based on their investibility and assign each stock to INV1, INV2 and INV3 groups as before. Then within each INV group, stocks are again sorted by their turnover ratio and assigned to each of small, medium and large size categories. This gives us 9 double sorted groups. Within each of the size-INV groups, we rank stocks by their ranking period returns and assign them to one of P1-P5 five portfolios, with P1 as losers and P5 as winners. Then we can construct our turnover-neutral IWF-based strategies: for each INV category, the winner portfolio consists of winner stocks (P5) stocks from each of the three turnover groups, while loser portfolio consists of loser stocks (P1) stocks also from each turnover group. And the momentum profits are the difference in returns between winner and loser portfolios. The t-statistics reported are computed using the Newey-West correction for heteroskedasticity and autocorrelation.

INV Groups	Test Period Returns						1990-1995	1996-2001
		P1	P2	P3	P4	P5	P5-P1	P5-P1
INV1	Mean	-0.0109	-0.0100	-0.0070	-0.0036	-0.0018	0.0091	0.0078
	<i>t</i>	-1.92	-2.39	-1.6	-0.73	-0.38	2.79	2.58
INV2	Mean	-0.0081	-0.0127	-0.0112	-0.0065	-0.0037	0.0044	-0.0083
	<i>t</i>	-0.95	-1.78	-1.7	-1.11	-0.64	0.77	1.78
INV3	Mean	-0.0125	-0.0093	-0.0005	0.0042	0.0075	0.0201	0.0134
	<i>t</i>	-1.6	-1.42	-0.1	0.79	1.17	2.95	2.4
F-statistics for INV1=INV3							5.5847	0.9056
<i>(P-value)</i>							0.0188	0.3429
								5.2051
								0.0240

Table 7 (Continued)
Risk Adjustment for IWF-based Momentum Strategies

C: Country-Neutral IWF-Based Momentum Strategies

Table 7 Panel C reports the country-neutral IWF-based momentum returns. The country-neutral IWF-based strategies are constructed as follows: at the end of each ranking period, we categorize stocks based on their investibility and assign each stock to INV1, INV2 and INV3 groups as before. Then within each INV group, stocks from the same country are again sorted by their ranking period returns and assign them to one of P1-P5 five portfolios, with P1 as losers and P5 as winners. The t-statistics reported are computed using the Newey-West correction for heteroskedasticity and autocorrelation.

INV Groups	Test Period Returns for 1990-2002						1990- 1995	1996- 2002
		P1	P2	P3	P4	P5	P5-P1	P5-P1
INV1	Mean	-0.0078	-0.0071	-0.0077	-0.0044	-0.0062	0.0016	0.0002
	<i>t</i>	-1.47	-1.59	-1.77	-0.99	-1.41	0.75	0.08
INV2	Mean	-0.0100	-0.0087	-0.0070	-0.0089	-0.0083	0.0017	-0.0015
	<i>t</i>	-1.41	-1.3	-1.1	-1.38	-1.31	0.66	-0.46
INV3	Mean	-0.0069	-0.0047	-0.0004	0.0002	0.0013	0.0082	0.0073
	<i>t</i>	-1.13	-0.77	-0.07	0.03	0.24	2.91	2.72
F-statistics for INV1=INV3							8.9337	7.5590
<i>(P-value)</i>							0.0030	0.0067

Table 8
IWF-Based Momentum Profits for Individual Countries

Table 8 reports the IWF-based momentum strategies for each of the 16 sample countries. N is the number of observations for each portfolio in calculating the average returns. The t-statistics reported are computed using the Newey-West correction for heteroskedasticity and autocorrelation.

	Test Period Returns (1990-2001)						
	INV1	N	INV2	N	INV3	N	INV3-INV1
Argentina <i>t-statistic</i>	-0.0512 -0.91	52	0.0252 2.12	33	0.0216 3.10	136	0.0728
Brazil <i>t-statistic</i>	-0.0009 -0.13	144	-0.0090 -0.87	138	0.0060 0.80	128	0.0070
Chile <i>t-statistic</i>	0.0098 1.50	108	0.0163 3.04	145	0.0018 0.27	71	-0.0081
Colombia <i>t-statistic</i>	-0.0057 -0.76	139	-0.0101 -1.21	58	0.0190 3.51	57	0.0247
Greece <i>t-statistic</i>	0.0094 0.75	58	-0.0006 -0.05	66	0.0133 2.65	145	0.0039
Indonesia <i>t-statistic</i>	-0.0118 -1.40	100	0.0064 1.30	137	0.0155 1.27	48	0.0273
Jordan <i>t-statistic</i>	0.0053 1.24	145	0.0080 1.71	78		0	
Korea <i>t-statistic</i>	-0.0071 -1.26	73	0.0023 0.44	121	0.0093 0.82	44	0.0164
Malaysia <i>t-statistic</i>	0.0019 0.16	49	0.0002 0.05	129	-0.0050 -1.07	141	-0.0069
Mexico <i>t-statistic</i>	0.0165 1.53	138	0.0069 0.42	57	0.0225 6.33	145	0.0060
Pakistan <i>t-statistic</i>	0.0002 0.05	145	-0.0083 -0.99	60	0.0017 0.19	77	0.0015
Philippines <i>t-statistic</i>	-0.0028 -0.37	144	0.0016 0.19	81	0.0132 1.16	79	0.0160
Taiwan <i>t-statistic</i>	-0.0343 -2.98	27	-0.0018 -0.38	133	-0.1267	3	-0.0924
Thailand <i>t-statistic</i>	0.0109 0.79	93	0.0136 2.19	145		0	
Turkey <i>t-statistic</i>	0.0318 1.20	28	-0.0105 -1.67	73	0.0001 0.01	142	-0.0317
Venezuela <i>t-statistic</i>	0.0111 1.12	102		0	0.0015 0.14	111	-0.0095
MEAN	-0.0011		0.0027		-0.0004		0.0019

Table 9
Market-Adjustment for IWF-Based Momentum Profits

Table 9 presents the market risk-adjusted returns of the IWF-based momentum strategies. The risk-adjusted returns (reported as CAPM- α) are estimated as follows: Firstly, for each momentum portfolio, at the end of each ranking period, a monthly $\hat{\beta}$ is estimated by regressing the portfolio's past 36 months' returns in excess of the monthly risk free returns (calculated as the monthly returns of the 30-day U.S Treasury Bill) on the excess return on the Morgan Stanley Capital International All Country World Index (MSCI-ACWI),

$$R_{i,t} - r_{f,t} = \alpha + \beta(R_{m,t} - r_{f,t}) + \varepsilon_{i,t} \quad t = -36, -35, \dots, -1$$

where $R_{i,t}$ is the return on the momentum portfolio (P5-P1), $r_{f,t}$ is the interest rate on 30-day Treasury Bill, and $R_{m,t}$ is the return on ACWI. Then the risk-adjusted returns (reported as CAPM- α) are calculated as:

$$\hat{\alpha}_t = (R_{i,t} - r_{f,t}) - \hat{\beta}(R_{m,t} - r_{f,t})$$

Panel A reports the raw and market-adjusted returns for the IWF-based momentum strategies on all the stocks. Panel B presents the raw and market-adjusted returns for the country-neutral IWF-based momentum strategies. The t-statistics reported are computed using the Newey-West correction for heteroskedasticity and autocorrelation.

INV Groups		A: All Stocks		B: Country-Neutral	
		Raw	CAPM- α	P5-P1	CAPM- α
INV1	Mean	0.0084	0.0064	0.0016	-0.0010
	<i>t-statistic</i>	2.57	1.94	0.75	-0.45
INV2	Mean	0.0041	0.0070	0.0017	0.0001
	<i>t-statistic</i>	0.67	0.87	0.66	0.04
INV3	Mean	0.0198	0.0158	0.0082	0.0040
	<i>t-statistic</i>	2.92	1.81	2.91	1.22
INV3-INV1		0.0114	0.0094	0.0066	0.0050
F-statistics for INV1=INV3		5.9792	5.9792	8.9337	3.8944
<i>(P-value)</i>		0.0151	0.0151	0.0030	0.0497

Table 10
Momentum Profits for Strategies with Various Horizons

Table 10 gives the momentum profits where ranking and test periods are 6, 12, 24 and 36 months separately. We also report the returns at different horizons after controlling for country, size and turnover ratio. The t-statistics reported are computed using the Newey-West correction for heteroskedasticity and autocorrelation.

		6/6	12/12	24/24	36/36
Unconditional	Mean	0.0117	-0.0365	-0.0074	-0.0081
	<i>t-statistic</i>	2.34	-1.74	-3.71	-4.34
Cty-neutral	Mean	0.0037	-0.0047	0.0001	-0.0009
	<i>t-statistic</i>	1.76	-0.69	0.11	-1.20
Size-neutral	Mean	0.0109	-0.0371	-0.0093	-0.0102
	<i>t-statistic</i>	2.30	-1.80	-4.94	-5.52
TO-neutral	Mean	0.0120	-0.0344	-0.0073	-0.0081
	<i>t-statistic</i>	2.46	-1.66	-3.73	-4.71

Table 11
Momentum Profits for IWF-Based Strategies with Various Horizons

A: Momentum Profits for IWF-Based Strategies on All Stocks

Table 11 Panel A gives the momentum profits of IWF-based strategies where ranking and test periods are 6, 12, 24 and 36 months separately. The t-statistics reported are computed using the Newey-West correction for heteroskedasticity and autocorrelation.

		6/6	12/12	24/24	36/36
INV1	Mean	0.0084	-0.0413	-0.0060	-0.0064
	<i>t-statistic</i>	2.57	-1.87	-2.51	-2.76
INV2	Mean	0.0041	-0.0398	0.0017	-0.0022
	<i>t-statistic</i>	0.67	-1.92	0.7	-0.94
INV3	Mean	0.0198	-0.0463	-0.0122	-0.0146
	<i>t-statistic</i>	2.92	-2.16	-5.63	-6.86
F-statistics for INV1=INV3		5.9792	0.0888	11.1135	24.8590
<i>(P-value)</i>		0.0151	0.7660	0.0010	0.0000

B: Country-Neutral Momentum Profits for IWF-Based Strategies

Table 11 Panel B gives the momentum profits of country-neutral IWF-based strategies where ranking and test periods are 6, 12, 24 and 36 months separately. The t-statistics reported are computed using the Newey-West correction for heteroskedasticity and autocorrelation.

		6/6	12/12	24/24	36/36
INV1	Mean	0.0016	0.0037	0.0050	0.0049
	<i>t-statistic</i>	0.75	0.39	2.62	2.39
INV2	Mean	0.0017	-0.0053	0.0008	-0.0021
	<i>t-statistic</i>	0.66	-0.69	0.46	-0.91
INV3	Mean	0.0082	0.0050	-0.0022	-0.0054
	<i>t-statistic</i>	2.91	0.65	-1.28	-5.38
F-statistics for INV1=INV3		8.9337	0.0393	29.3509	71.5264
<i>(P-value)</i>		0.0030	0.8430	0.0000	0.0000

Table 12 Foreign Ownership Restriction and Price Efficiency

This table investigates the overreaction hypothesis. We limit our sample stocks to those that are consistently totally open or close to foreign investors, that is, stocks that have INV=1 or INV=0 for our whole sample period. Then each year, we run the market model regression for each stock, and calculate the average R^2 across stocks within the same INV level. To increase the power of the test, we use weekly returns and extend the sample period to the earliest date that the data is available for individual stock from the database. Test results are reported in the left panel of Table 12. As a robustness test, we change our cutting points and define highly investible stocks as stocks with average INV over the sample period greater than or equal to 0.8, and non-investible stocks as those with average INV less than or equal to 0.2. We rerun the test and report the results in the right panel of Table 12.

Test Period (12/1988-08/2002) Weekly data								
		#stocks	R-Square	Std.Dev		#stocks	R-Square	Std.Dev
Argentina	INV=1	6	0.3477	0.2242	INV>=0.8	27	0.5034	0.2284
	INV=0	3	0.5081	0.1510	INV<=0.2	7	0.4075	0.1461
Brazil	INV=1	6	0.1362	0.2100	INV>=0.8	59	0.2765	0.1774
	INV=0	15	0.0896	0.0728	INV<=0.2	29	0.1378	0.1124
Chile	INV=1	1	0.3366		INV>=0.8	8	0.3272	0.1683
	INV=0	8	0.1071	0.0900	INV<=0.2	12	0.1146	0.0817
Colombia	INV=1	1	0.2362		INV>=0.8	3	0.2072	0.1668
	INV=0	18	0.0924	0.0918	INV<=0.2	23	0.1095	0.0993
Greece	INV=1	13	0.2760	0.1700	INV>=0.8	53	0.2871	0.1255
	INV=0	7	0.0802	0.0757	INV<=0.2	11	0.1248	0.1168
Indonesia	INV=1	1	0.4067		INV>=0.8	5	0.2805	0.1084
	INV=0	65	0.1007	0.1212	INV<=0.2	74	0.1115	0.1200
Jordan	INV=1	0			INV>=0.8	0		
	INV=0	59	0.0947	0.0797	INV<=0.2	72	0.1016	0.0794
Korea	INV=1	6	0.2666	0.1904	INV>=0.8	48	0.2693	0.0949
	INV=0	4	0.1228	0.0509	INV<=0.2	53	0.2088	0.1098
Malaysia	INV=1	38	0.3772	0.1352	INV>=0.8	132	0.4167	0.1310
	INV=0	9	0.1736	0.1182	INV<=0.2	9	0.1736	0.1182
Mexico	INV=1	22	0.2763	0.1970	INV>=0.8	74	0.3056	0.2209
	INV=0	33	0.1011	0.1201	INV<=0.2	36	0.1012	0.1179
Pakistan	INV=1	0			INV>=0.8	4	0.5303	0.1104
	INV=0	70	0.1330	0.1133	INV<=0.2	78	0.1458	0.1144
Philippines	INV=1	0			INV>=0.8	7	0.3218	0.1333
	INV=0	35	0.1853	0.1596	INV<=0.2	51	0.1731	0.1480
Taiwan	INV=1	0			INV>=0.8	0		
	INV=0	6	0.4701	0.0580	INV<=0.2	46	0.4302	0.1347
Thailand	INV=1	0			INV>=0.8	1	0.2023	
	INV=0	13	0.2533	0.1378	INV<=0.2	41	0.2634	0.1325
Turkey	INV=1	12	0.3343	0.1917	INV>=0.8	45	0.4001	0.1455
	INV=0	1	0.3603		INV<=0.2	2	0.3537	0.0094
Venezuela	INV=1	0			INV>=0.8	4	0.4688	0.1180
	INV=0	10	0.1923	0.3052	INV<=0.2	11	0.1885	0.2898
Sum	INV=1	106	0.2952	0.1912	INV>=0.8	470	0.3413	0.1486
	INV=0	356	0.1905	0.1193	INV<=0.2	555	0.1955	0.1227